

A Comparative Percolation Analysis of the Internet AS Topology and the World Wide Web

Work Plan Final Project, Modelling Real World Problems (900393SCIY)

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1. Study topic and context

Modern society depends on two large-scale information infrastructures: the Internet at the level of Autonomous Systems, and the World Wide Web at the level of hyperlinked pages. Both are commonly described in the network-science literature as scale-free networks, meaning that their degree distributions are dominated by a small number of highly connected hubs. A central qualitative claim about these networks is that they are simultaneously robust to random failure and fragile under targeted attack.

Although this property has been demonstrated theoretically for the Barabási–Albert (BA) model and is supported by classical empirical work, it remains an open question to what extent the two infrastructures behave identically in practice. The AS-level Internet and the hyperlink Web differ in how they grow, in their typical degrees, and in higher-order structural features such as clustering. A direct, controlled comparison of how each system behaves under random failure and targeted attack is therefore informative both as a test of theory and as input to discussions about infrastructure resilience and cybersecurity policy.

2. Research question

In this research, the effect of node removal strategy (random failure versus degree-targeted attack) on the size of the giant connected component in the Internet AS-level topology and the WWW hyperlink network will be investigated, and the empirical results will be compared with the predictions of equivalent Erdős–Rényi (ER) and Barabási–Albert (BA) synthetic networks.

Two more specific sub-questions follow from this general formulation:

- Do both real networks exhibit the asymmetry between random failure and targeted attack predicted for scale-free graphs, and is the asymmetry the same for the two systems?
- Is the heavy-tailed degree distribution alone sufficient to reproduce the observed fragility, or do higher-order structural features (clustering, degree–degree correlations) contribute?

3. Model description

3.1 Basic model

The basic model is a percolation process on an undirected, unweighted graph. Starting from the full network, nodes are removed one at a time according to a chosen ordering until a fraction of the nodes has been removed. After each removal step, the size of the largest connected component, normalized by the original number of nodes, is recorded. Two removal orderings are considered: a random ordering, corresponding to uniform random failure, and a descending-degree ordering, corresponding to a targeted attack on hubs. The percolation threshold is identified as the fraction of removed nodes at which the giant component collapses.

Four networks are studied within the same framework:

- **AS-level Internet topology:** the Oregon-1 snapshot from the SNAP repository ($\sim 10^4$ nodes), where nodes are Autonomous Systems and edges are observed BGP peering relationships.

- **WWW hyperlink network:** the web-NotreDame graph from SNAP, treated as undirected, where nodes are pages within the nd.edu domain and edges are hyperlinks. This is the dataset originally used to identify scale-free structure of the Web.
- **ER null model:** an Erdős–Rényi random graph with the same number of nodes and the same average degree as each real network.
- **BA null model:** a Barabási–Albert preferential-attachment graph with parameters chosen to match the average degree of each real network.

3.2 Potential extensions

4. Data collection and in silico experiments

4.1 Data

All network data are obtained from the Stanford Large Network Dataset Collection (SNAP, <https://snap.stanford.edu/data/>).

4.2 Outcome variables

The primary outcome variable is the relative size of the largest connected component. The percolation threshold is defined operationally as the value at which the largest component first drops below a small threshold (1% of N).

4.3 Experimental design

4.4 Implementation

All experiments will be implemented in Python using the NetworkX library for graph manipulation, the powerlaw library for degree-distribution fitting, and Matplotlib for figures.

5. Answering the research question

The proposed design directly addresses the research question. The contrast between random and targeted removal on each real network operationalizes the "robust yet fragile" claim and provides a quantitative comparison between the two infrastructures. The ER and BA nulls serve as theoretical baselines: agreement with the BA prediction supports the standard scale-free explanation, while systematic deviations point to features beyond the degree distribution. The optional configuration-model extension further isolates the degree sequence as an explanatory factor. The combination of these conditions therefore allows the work to move from a descriptive comparison to a structured causal account, while remaining within the methods covered in the course.

6. Feasibility and timeline

The project is feasible within the time available. The networks selected are at the upper end of what runs comfortably on a laptop.

7. Literature

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